

Ammar Asad

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Education

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelors in Electrical Engineering

2026 – Present

• Current CGPA: **3.82 / 4.00**

• Elected Class Representative (CR) for the EE Cohort

FSc Pre-Engineering

Score: 93%

Matriculation

Score: 96%

Technical Toolbelt

- **Programming Languages:** C++, Python, MATLAB
- **Hardware & Systems:** Arduino IDE, ESP32, Microcontrollers
- **Design & Simulation Software:** AutoCAD, LTspice, Simulink
- **Version Control & Tools:** Git, GitHub, L^AT_EX, Markdown

Engineering Projects

BoDePlotter Pro — Automated Testing & Instrumentation Tool

C++, Hardware Automation

- Engineered an automated hardware-software system in C++ to trace physical circuit frequency responses dynamically, completely replacing traditional manual oscilloscope and signal generator tracking sequences.
- Implemented concurrent polynomial root-finding routines using the **Durand-Kerner** (**Durand-Kerner** / **Aberth-Ehrlich**) algorithm inside execution loops to isolate network transfer functions simultaneously.
- Integrated **Horner's Method** to accelerate floating-point polynomial evaluation metrics, minimizing processing overhead during real-time complex stream conversions.

PyDIPS — Machine Learning Network Security Suite

Python, Machine Learning, Cybersecurity

- Developed an end-to-end Python-based Digital Intrusion Prevention Suite (PyDIPS) establishing a localized Intrusion Detection and Prevention System (IDS/IPS) pipeline.
- Authored low-level socket-sniffing scripts to harvest packet features into a local data logging network while writing a multi-threaded network attack simulator to validate system response limits.
- Trained an artificial intelligence classifier package (‘.pkl’) over raw CSV traffic history features to dynamically isolate and intercept malicious network packets and threat vectors in real time.

Industrial Solar Farm Layout & Design

AutoCAD, Infrastructure Planning

- Drafted comprehensive structural footprints, access paths, and spatial array layouts for multi-megawatt photovoltaic installations using AutoCAD, ensuring complete link compliance with industrial design criteria.
- Calculated pitch angles and inter-row spacing constraints to completely mitigate localized winter solstice shading losses, optimizing annual solar irradiance tracking yields.
- Mapped detailed single-line electrical routing plans spanning string combinations, central inverter room layouts, a 1000kVA step-up transformer substation configuration, and high-capacity Battery Energy Storage System (BESS) racks.

Automatic Power Factor Correction (APFC) Meter

Embedded C++, Smart Grid Systems

- Designed and fabricated an automated smart grid hardware monitoring unit leveraging microcontroller intelligence to optimize power network loading efficiency.
- Authored real-time firmware loops to track current-voltage phase deltas, computing instantaneous reactive power overheads across volatile inductive loads.
- Deployed automated relay logic to actuate localized capacitor steps, correcting lagging power factors from ~0.72 up to an efficient target benchmark of 0.95.

Academic & National Honors

- **Pride of Pakistan Nominee (2025):** Nominated by the Government of Pakistan for Independence Day Honors in recognition of exceptional dedication to the sciences and representing the nation globally.
- **Highest Ranked Silver Medalist – INSO (2025):** Ranked **9th globally** at the International Nuclear Science Olympiad; selected for the national team after a multi-stage vetting process and secured the sole international silver medal for Pakistan.
- **Pakistan’s Mathematics Olympiad (2024):** Attained an elite national rank of **23rd** in the nationwide high school mathematics challenge jointly organized by FBISE and LUMS.
- **Copernicus Physics Olympiad (2023):** Awarded an international **Honorable Mention** for theoretical physics problem-solving benchmarks.
- **Open Cybersecurity Challenge:** Achieved National Semi-Finalist standing in the university-level security infrastructure tracking competition.

Leadership & Industry Experiences

Elected Class Representative (CR) — NUST SEECS 2026 – Present

- Act as the primary administrative and operational link between university department heads, faculty mentors, and an undergraduate Electrical Engineering cohort; organize academic tracking groups and coordinate departmental operations.

IEEE NUST Student Branch — Member, Publications Department 2026 – Present

- Curate, draft, and technically review academic content, research summaries, and engineering publications tailored for the electrical engineering and automation student body.

STEM Youth Lead — PIEAS Collaboration 2025

- Served as a core public liaison to foster scientific youth engagement across specialized STEM disciplines, organizing outreach frameworks in collaboration with institute mentors.

Autonomous Drone Systems Training — Rawalpindi Technology Park Extracurricular

- Led a high school team at the 18th edition of the National Design Build and Fly Competition (NDBFC) held at GIK Institute; completed practical systems training in autonomous drone build configurations and earned formal certification.